NetLift: A Causal Decision System for Incremental Profit Optimization

Phase 2: Literature Review

Digital Egypt Pioneers Initiative (DEPI), Microsoft Machine Learning Engineer Track

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This review covers the sources behind the project's experimental design and its first family of uplift models: how the two datasets were produced and what they can support, how non-compliance changes what can be estimated, and the meta-learners and class transformation that the baseline uplift models are built on. Each source is summarised, followed by what it means for this project.

1. Randomized experiments, non-compliance and the two datasets

Author: Aliaa Gomaa · Task: T25 · Reviewed by: Mazen

1. Diemert et al. (2021), the Criteo uplift benchmark

Citation. Diemert, E., Betlei, A., Renaudin, C., Amini, M.-R., Gregoir, T., & Rahier, T. (2021). *A Large Scale Benchmark for Individual Treatment Effect Prediction and Uplift Modeling*. arXiv preprint arXiv:2111.10106. https://arxiv.org/abs/2111.10106

Summary. Public uplift datasets were small, so the authors release about 14 million users from advertising incrementality tests at Criteo, where a random share of users is held out from ads. Each row holds twelve anonymised features measured before assignment, a treatment flag, an exposure flag (did the user actually see an ad), and visit and conversion outcomes over two weeks. Because the tests used different treatment shares, the test identity acted as a hidden confounder in the 2018 release, so every test was subsampled to one common ratio of about 85% treated. The authors then check with a classifier two-sample test that treatment cannot be predicted from the features, and that the features do predict the outcomes. Their main result is that baseline uplift methods whose confidence intervals overlap on small samples only become separable at around one million test points. The paper also states that control users are, by definition, never exposed.

What NetLift takes from it. It fixes which release of Criteo we use: the 13,979,592 rows counted in T16 are the figure in Table 1 of this paper, whereas the 2018 paper cited in `docs/unit-of-observation.md` describes the earlier release, later decommissioned, with 25.3 million rows. Its statement that control users are never exposed matches what T17 counted in the data and what Section 6 now says: non-compliance is one-sided. Its classifier two-sample test, which asks whether treatment can be predicted from the features, is the same idea as the treatment-predictability AUC in the specification's Randomization Integrity Check. Finally, the test identifier was never released, so no true group id exists for the sensitivity check in issue #29, which has to use a feature-profile proxy.

2. Hillstrom (2008), the MineThatData e-mail challenge

Citation. Hillstrom, K. (2008). *The MineThatData E-Mail Analytics And Data Mining Challenge*. MineThatData blog, 20 March 2008. https://blog.minethatdata.com/2008/03/minethatdata-e-mail-analytics-and-data.html

Summary. Hillstrom, an analytics consultant, posted a dataset from an e-mail test and asked readers to analyse it. 64,000 customers who had purchased in the last twelve months were randomly split into thirds: a Mens e-mail, a Womens e-mail, or no e-mail. Visits, conversions and spend were tracked for two weeks afterwards. The pre-campaign attributes are recency, past spend, whether the customer bought Mens or Womens items, an urban/suburban/rural zip class, a new-customer flag and purchase channel. The post is a dataset description and a set of questions, not a study, so it reports no result and says nothing about whether e-mails were delivered or opened.

What NetLift takes from it. The variable list has no exposure or delivery column, only the arm assigned, so Hillstrom supports intention-to-treat analysis only, and the complier (ToT) extension discussed in Section 6 applies to Criteo alone. The published size also depends on the contrast: 64,000 is the full three-arm file, while the Criteo papers list Hillstrom as 42,693 rows, which the 2018 paper identifies as the Womens-versus-no-e-mail split. `scripts/figures/response_by_arm.py` keeps all three arms, so it matches the full file; any two-arm analysis added later must say which contrast it uses.

3. Angrist, Imbens & Rubin (1996), instrumental variables and compliers

Citation. Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996). *Identification of causal effects using instrumental variables*. Journal of the American Statistical Association, 91(434), 444-455. https://doi.org/10.2307/2291629

Summary. The paper treats studies where assignment to a treatment is random but compliance is imperfect, so actual receipt of treatment is not random. Comparing people by assignment, the intention-to-treat analysis, remains valid but answers a question about being offered the treatment. Economists used instrumental variables for this problem, but under models with constant treatment effects. The authors embed the IV approach in the Rubin Causal Model and show that, under a few interpretable assumptions, the IV estimand is the average effect for compliers only; without those assumptions it is just a ratio of intention-to-treat effects with no causal reading. They illustrate it with the Vietnam-era draft lottery and test sensitivity to the assumptions.

What NetLift takes from it. Section 6 of the specification and `docs/unit-of-observation.md` decide that the MVP estimates ITT only, with the Wald-ratio ToT among compliers as an optional Advanced-tier extension. This paper is the reason for that split: the ITT is always identified because assignment is random, while the complier effect is a causal average only under extra assumptions, above all the exclusion restriction that Section 6 singles out. It also explains why `scripts/criteo_complier_baseline.py` reports the ITT next to the Wald ratio, and why the ToT is described as an effect among compliers (about 3.6% of treated users were exposed), never as an effect on everyone treated.

How these sources relate

Hillstrom is the clean case: randomized assignment and no way to observe compliance, so ITT is the only estimand. Criteo adds exposure and one-sided non-compliance, which is exactly the setting Angrist, Imbens and Rubin analyse. Diemert et al. connect the two, since their dataset is presented as a larger successor to Hillstrom's. Together they give NetLift one rule: ITT on both datasets, a complier effect only where exposure exists.

2. Meta-learners and class transformation

Author: Mohammed Sherif · Task: T26

1. Künzel, Sekhon, Bickel & Yu (2019): the S-, T- and X-learners

Citation. Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). *Metalearners for estimating heterogeneous treatment effects using machine learning*. Proceedings of the National Academy of Sciences, 116(10), 4156–4165. https://doi.org/10.1073/pnas.1804597116 (preprint: https://arxiv.org/abs/1706.03461)

What the paper says. Any base model can be turned into a treatment-effect estimator by wrapping it in a "meta-learner" instead of redesigning it. The paper formalises the S-learner (one model, treatment as an input) and the T-learner (one model per arm), then proposes the X-learner for groups of very different sizes: it imputes each unit's missing counterfactual from the opposite arm's model, fits an effect model on each side's imputed effects, and combines them with a propensity-score weight. Under stated regularity conditions the X-learner can reach a near-parametric rate, and it is competitive in simulation, though the authors stress that no meta-learner is best everywhere.

What NetLift takes from it. These are the MVP's two mandatory estimators, T-learner and X-learner (Part I of `docs/NetLift_Project_Specification.tex`, Final Recommended Product → MVP), and the basis for making X-learner primary: Criteo's 0.85 treated share leaves a thin control arm, the situation the X-learner was designed for. That mapping is the project's own extrapolation. The paper's evidence comes from its own simulations and field experiments, not from Criteo or Hillstrom, so it supports the choice without proving it.

2. Jaśkowski & Jaroszewicz (2012): class variable transformation

Citation. Jaśkowski, M., & Jaroszewicz, S. (2012). *Uplift modeling for clinical trial data*. ICML 2012 Workshop on Machine Learning for Clinical Data Analysis, Edinburgh, Scotland. https://home.ipipan.waw.pl/sj/pdf/logsemi.pdf

What the paper says. A classifier trained on outcomes alone says who responds, not who responds because of treatment, so the authors relabel the data. They define $Z = 1$ for treated patients with a positive outcome and control patients with a negative one, and show that uplift equals $2P(Z=1|X) - 1$ when treatment and control are equally likely and the treatment indicator G is independent of X (as in randomised assignment). For unbalanced data they propose reweighting or resampling so that the assumption becomes valid, and give a short argument: writing $P(Y,X,G) = P(Y,X|G)P(G)$ shows that reweighting affects only $P(G)$, not the conditional distribution. On two clinical trial datasets (bone marrow transplant, tamoxifen) and one non-trial dataset (UCI hepatitis, which the paper itself describes as "not a controlled clinical trial data"), the ranking of methods changes: the double-model approach is best on bone marrow transplant and tamoxifen (where the transformation is second and last of four respectively, by area under the uplift curve), while the transformation beats both two-separate-classifier variants on hepatitis, though a semi-supervised variant beats it there too.

What NetLift takes from it. This is the source of the "class-transformation trap" (Part II): the identity needs equal assignment probabilities, and Criteo's treated share is 0.85 (11,882,655 of 13,979,592 rows, the counted figure in the spec's Criteo-UPLIFT dataset table and `docs/unit-of-observation.md`), so the naive version is distorted there. The paper supplies the reweighting argument; the spec's rule that the method may be used only in its propensity-weighted form, and that the naive version is documented as a trap, is NetLift's own requirement built on it. The method is also written for one treatment against one control; for Hillstrom's two email arms, the spec's MVP analyses one arm at a time against the shared control, and joint multi-arm modelling is Advanced tier.

3. Nie & Wager (2021): the R-learner

Citation. Nie, X., & Wager, S. (2021). *Quasi-oracle estimation of heterogeneous treatment effects*. Biometrika, 108(2), 299–319. https://doi.org/10.1093/biomet/asaa076 (preprint: https://arxiv.org/abs/1712.04912)

What the paper says. The standard S- and T-learners mainly estimate outcome functions and take the CATE as a difference of fitted outcomes. They do not necessarily need a propensity model, and the loss does not target the effect directly, so bias in the outcome fits can pass into the estimate. The R-learner instead cross-fits two nuisance estimates, the conditional mean outcome and the propensity score, uses them to build a residualised objective via Robinson's transformation, and minimises that "R-loss" over candidate effect functions. For penalised kernel regression, the paper shows the error rate matches an oracle that knows the nuisance functions ("quasi-oracle"), provided those are estimated at a stated minimum rate.

What NetLift takes from it. The spec lists the R-learner as Advanced tier, "if time allows" (Part II, Modeling Approaches), and requires out-of-fold nuisance models: incorrect cross-fitting "reintroduces exactly the bias the estimator exists to remove, and the error is invisible in the metrics" (Part III, Splitting Strategy, which applies this to the X- and R-learners). Whether NetLift's base learners meet the paper's rate condition on Criteo's anonymised features is unverified, which is why this stays Advanced.

4. Kennedy (2023): the DR-Learner

Citation. Kennedy, E. H. (2023). *Towards optimal doubly robust estimation of heterogeneous causal effects*. Electronic Journal of Statistics, 17(2), 3008–3049. https://doi.org/10.1214/23-EJS2157 (preprint: https://arxiv.org/abs/2004.14497)

What the paper says. Kennedy studies a two-stage doubly robust estimator that he names the DR-Learner, following the naming scheme from Nie & Wager. The gap he addresses is that it was unclear when such estimators are optimal, not just consistent, once the true effect has structure such as smoothness or sparsity. Theorem 2 gives a general, model-free error bound whose doubly-robust form means the error depends on the nuisance errors jointly, so oracle efficiency needs weaker conditions than earlier results. A local-polynomial variant (Section 5, the lp-R-Learner) reaches oracle efficiency under weaker conditions still, with specialised sample splitting.

What NetLift takes from it. The spec lists the DR-learner as Advanced tier, alongside the R-learner, and describes it as staying consistent if either the outcome or the propensity model is correct, at the price that "Three-way cross-fitting costs sample efficiency and is easy to implement wrongly" (Part II, Modeling Approaches). That consistency description is the standard doubly-robust property; the paper's own stated contribution is the error-bound and efficiency analysis built on top of it, so the review cites it for the rates, not as the source of the property itself.

How these sources relate

Künzel et al. frame the family the other three sit inside: S-, T- and X-learners wrap off-the-shelf models, while class transformation is a single-model route that behaves oppositely with respect to group balance. The X-learner is designed to help when treated and control groups are very unequal, whereas the transformation identity as stated needs equal assignment probabilities, so an imbalanced split like Criteo's breaks it. Nie & Wager and Kennedy both rely on estimated nuisance functions (outcome and propensity) instead of differencing outcome fits alone, the R-learner through a residualised loss and the DR-Learner through a two-stage doubly robust construction. Each paper proves its guarantee only under its own conditions and none tests NetLift's datasets directly, nor Hillstrom's three-arm structure specifically, Hillstrom is out of scope for all four papers here, and enters NetLift only through the spec's own one-arm-at-a-time MVP rule. So the choices of X-learner for the MVP and R- and DR-learners for Advanced are informed extrapolations, which the spec's conditional wording ("if time allows", "Advanced tier") already reflects.